

# 6MARK027C

# Digital Marketing, Social Media & Web Analytics

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## WEEK 2:

## INFORMATION RETRIEVAL SYSTEMS AND SEARCH ENGINES

# Week 1

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## *Recap from last week*

- ❖ Purpose of marketing
- ❖ Marketing principles (3Ps, 4Cs)
- ❖ Traditional vs. digital marketing
- ❖ Two proposed marketing models
  - ❖ Ash (2012) and Miller (2012)
- ❖ Acquisition (suspects => prospects)

# Week 2

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## *Plan for today*

- ❖ Principles of the web
- ❖ Information and data retrieval systems (IR, DR)
  - ❖ Retrieval models
- ❖ Search engines (SE)



# The early web

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- ❖ **Tim Bernes-Lee** is generally accreted with the invention of the world wide web (www) *circa* 1989
- ❖ Born in South West London
- ❖ Software engineer
  - ❖ background in telecommunications
- ❖ Consultant for CERN
  - ❖ European Organization for Nuclear Research, Geneva



# The early web

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- ❖ **Scientists would travel to Geneva to carry out large experiments at CERN**
- ❖ Lots of data/reports generated
  - ❖ Desire to have that data available to their organisation when they went back home
- ❖ People wanted to read about the experiments performed by others
  - ❖ Collaborate/share ideas



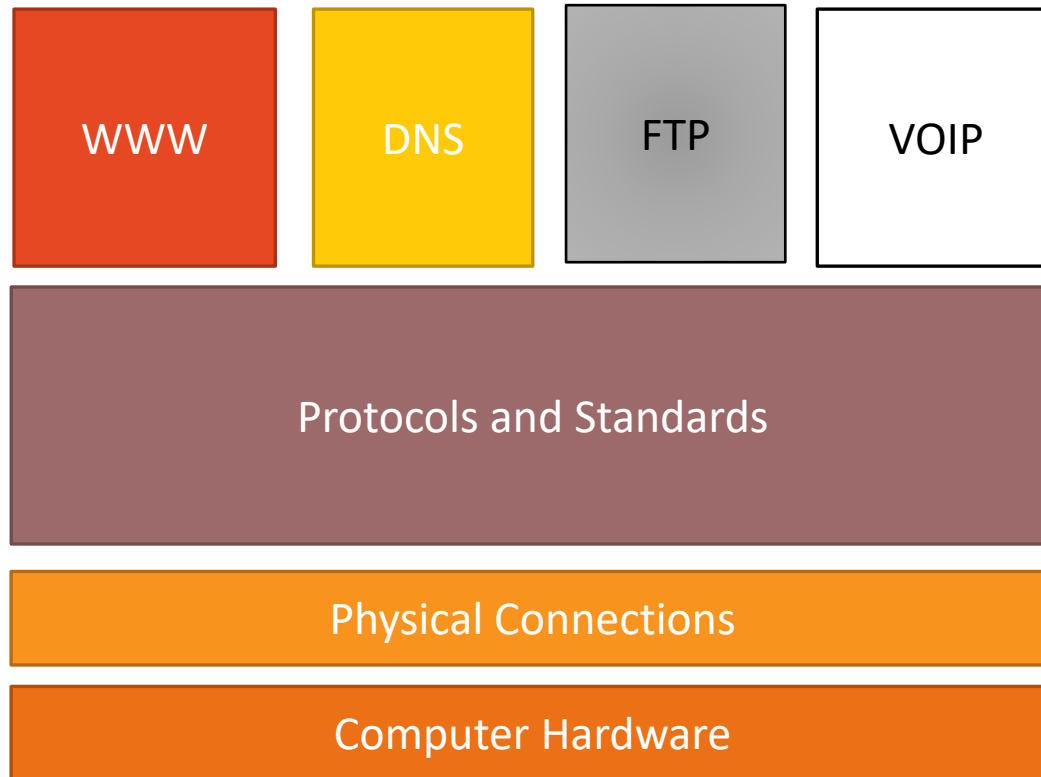
# The early web

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- ❖ While consulting for CERN he put out a plan to exchange data using links between the various documents
  - ❖ Easier for researchers to see what has already been done
- ❖ Hypertext (HTML), URI, HTTP
- ❖ Ability to view this information regardless of the type of computer you were using
- ❖ Tim's ideas were not initially accepted
- ❖ First web page viewed in 1990

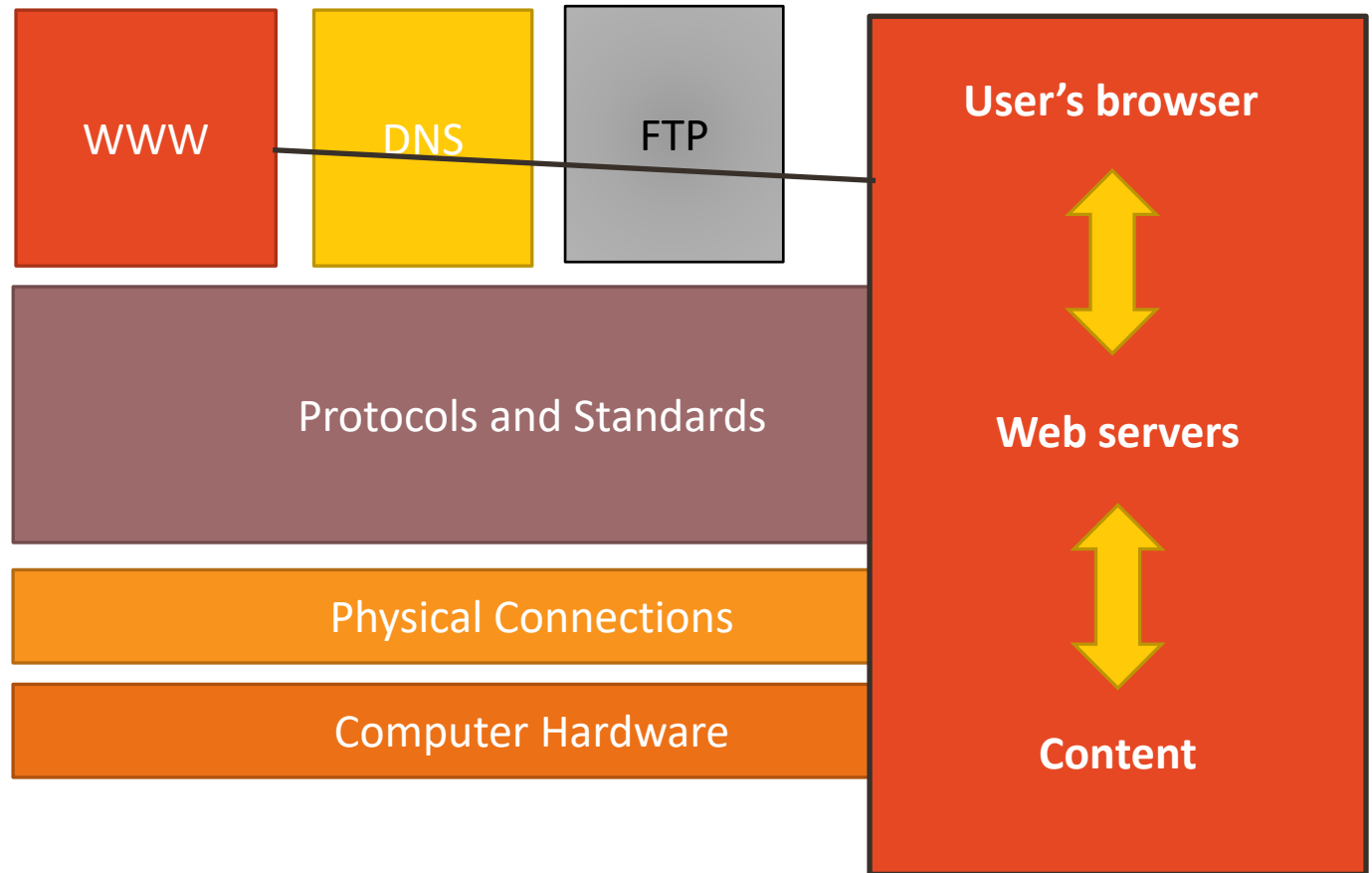
# The modern internet

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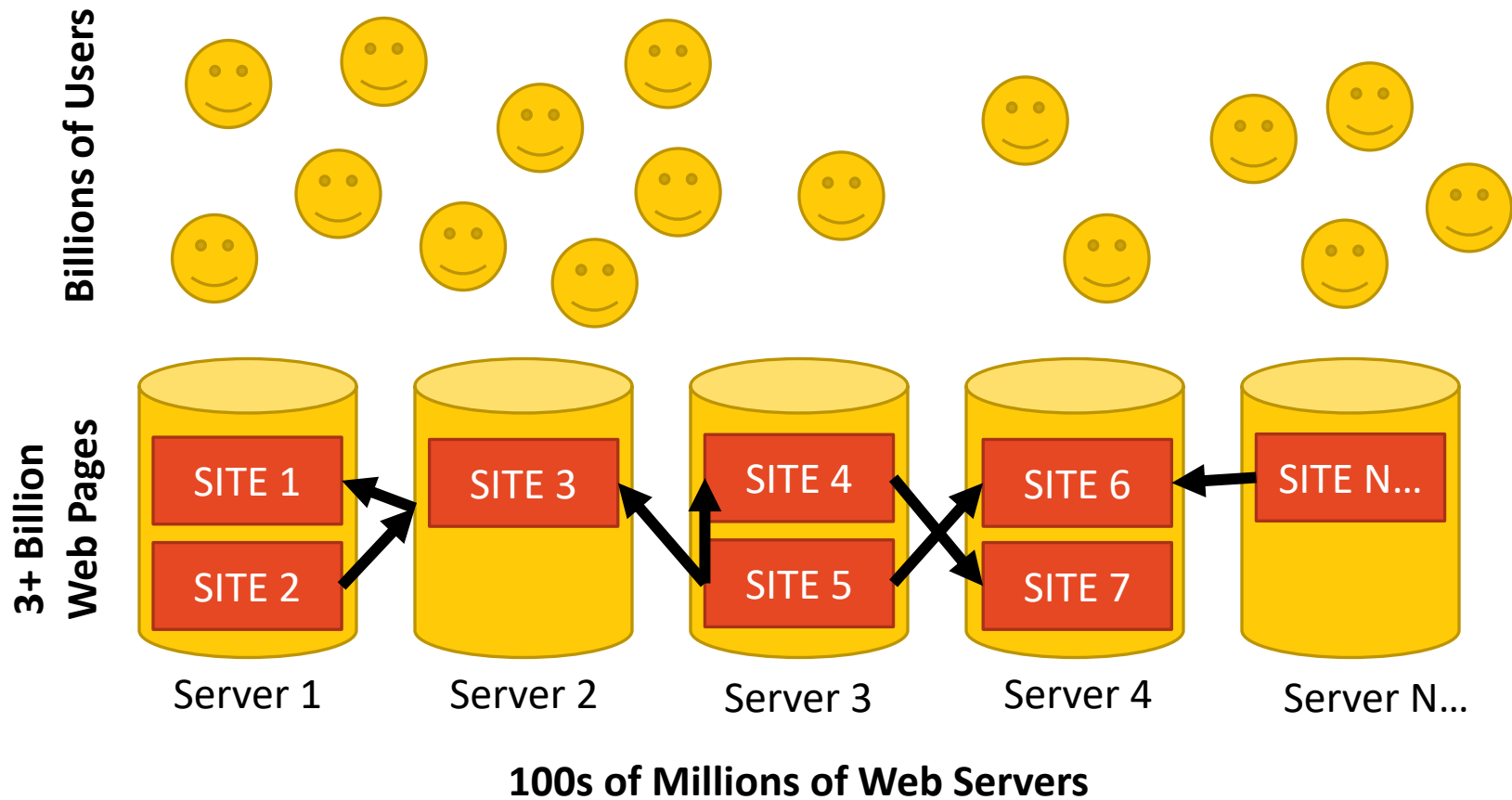
**The web (WW) is just one use case of the internet**

# The modern internet





# Layout of the web



# Locating relevant content

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- ❖ In earlier versions of the web, the primary way to locate relevant information was through hyperlinks (PAGE A => link => PAGE B).
- ❖ As the web grew, alternative methods of using web directories were proposed. Each maintainer of a directory curated links to relevant sites in their niche.
- ❖ A longer term solution was needed.
- ❖ Throughout the 1990s and early 2000s several different search engines entered the market using ideas from information retrieval (IR) theory.

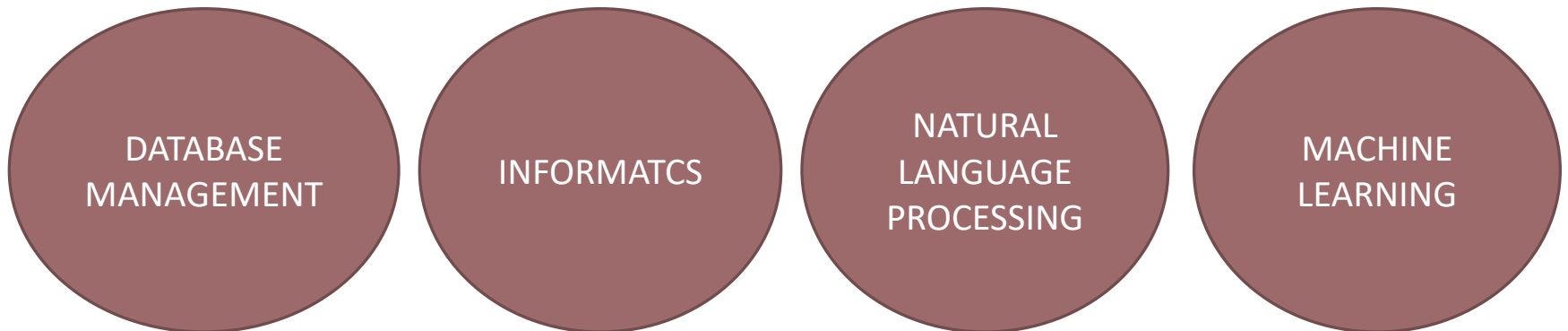
# Information Retrieval

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## Information retrieval (IR) systems

Any system that enables users to obtain information (**relevant documents or fragments**) relevant to an information need (**query**) from a document collection (**corpus**). Searches can be based on metadata or on full-text (or other content-based) indexing.

IR Systems and theory involve a number of specialist scientific disciplines



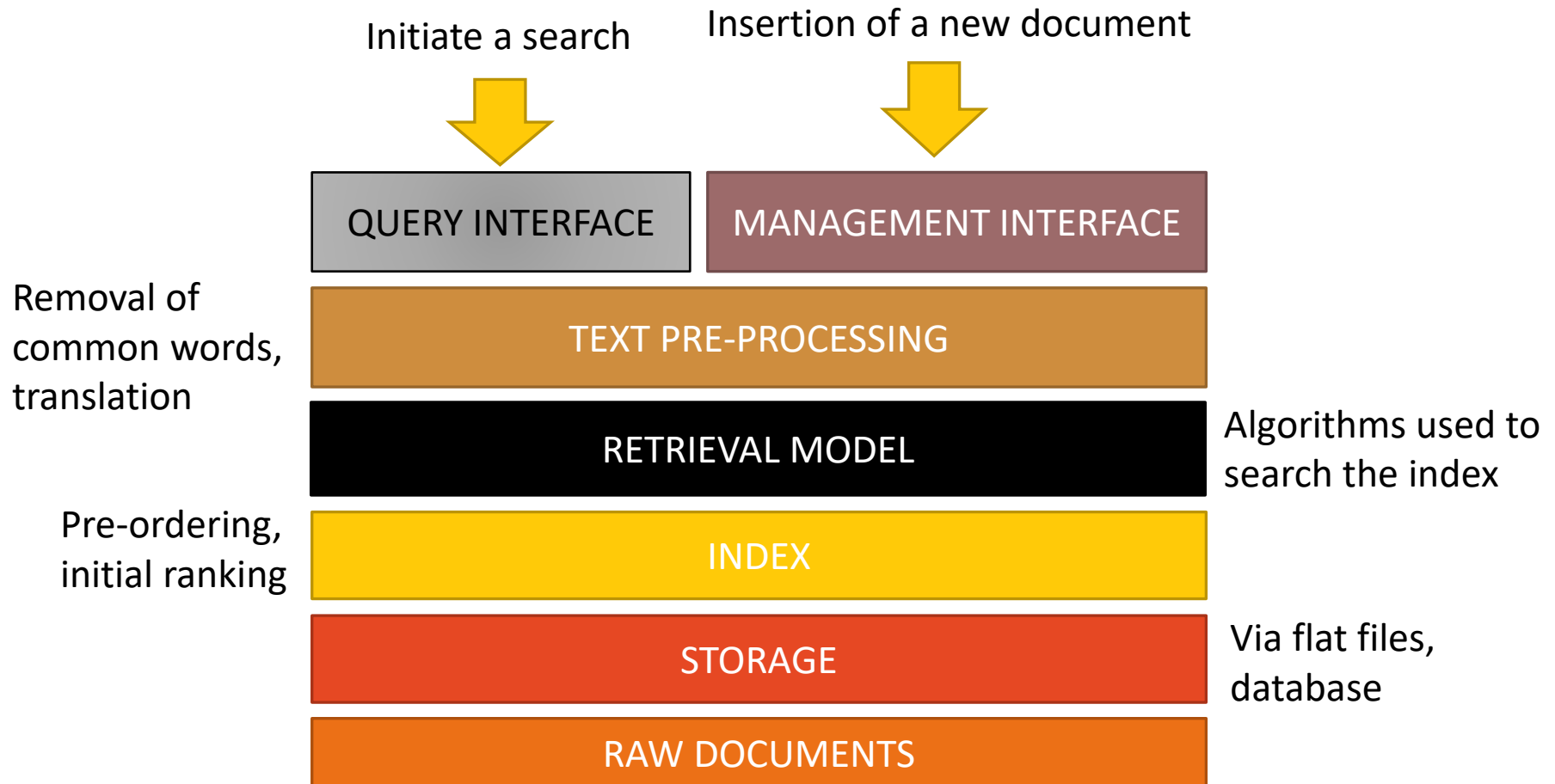
# Data vs Information retrieval

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|                  | Data Retrieval                        | Information Retrieval                                   |
|------------------|---------------------------------------|---------------------------------------------------------|
| What we retrieve | Structured Data / Clear Semantics     | Mostly unstructured / free text with metadata           |
| Queries          | Formally defined queries/ unambiguous | Vague / ambiguous / often expressed in Natural Language |
| Results          | Exact                                 | Sometimes relevant – others not relevant                |
| Interaction      | Ad hoc, reporting                     | Significant interaction, research, refinement of query  |

# Architecture of an IR system

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# The indexing component

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## RAW DOCUMENT

#1 “The hotel is located next to the city railway station.

## RAW DOCUMENT

#2 The Plaza hotel is a 2 min walk from the coastline.

## INDEX

**Organised  
Alphabetically**

| TERM      | DOCUMENT | POSITION |
|-----------|----------|----------|
| city      | #1       | 8        |
| coastline | #2       | 11       |
| from      | #2       | 9        |
| hotel     | #1       | 2        |
| hotel     | #2       | 3        |

The term “a” doesn’t appear in the index even though it would be placed before “city”. **WHY?**

# The text pre-processing component

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## TEXT PRE-PROCESSING

- ❖ Transforms USER queries before they are checked against the INDEX
- ❖ Modifies the TERMS placed into the index when a new DOCUMENT is added to the collection.

## WHY?

- ❖ Which documents would the TERM “City” match?
- ❖ How about the TERMS “London”, “Madrid” or “hotels”?
- ❖ What we do we know about the TERMS “a”, “and”, “to”, “in”.

# Text pre-processing strategies

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## TEXT PRE-PROCESSING

- ❖ Tokenisation: break text into individual TERMS (keywords)
- ❖ Stem TERMS to 'root' words e.g. Computer -> comput, running -> ran
- ❖ Remove common 'stopwords' e.g. "a", "the", "it", "to"
- ❖ Strip unwanted characters/markup e.g. HTML tags, punctuation, numbers, etc.
- ❖ Detect common phrases and named entities - possibly by using a domain specific dictionary



# Retrieval models and algorithms

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## RETRIEVAL MODEL

When a QUERY is conducted, documents in the collection will fall into one of **FOUR** possible categories.

- 1) Relevant documents that are selected
- 2) Relevant documents that are not selected
- 3) Non-Relevant documents that are selected
- 4) Non-Relevant documents that are not selected.

It is the responsibility of the retrieval model (algorithm) to search the INDEX to identify relevant documents and determine a possible ranking.

# Retrieval models and algorithms

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## RETRIEVAL MODEL

$$Precision = \frac{\text{relevant and selected}}{\text{relevant and selected} + \text{non-relevant and selected}}$$

$$Recall = \frac{\text{relevant and selected}}{\text{relevant and selected} + \text{relevant and not selected}}$$

Precision relates to the retrieval models ability to deliver a high proportion of results that the user will find useful.

Recall reflects the ability of the retrieval model to correctly identify relevant documents in the collection.

# Potential algorithms

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## RETRIEVAL MODEL

### ❖ Standard Boolean Model

- ❖ Documents are selected and retrieved if the search TERM appears in a document.

### ❖ Term-Frequency Model

- ❖ Documents with the highest frequencies of the search TERM are selected.

### ❖ Term-Frequency-Density Model

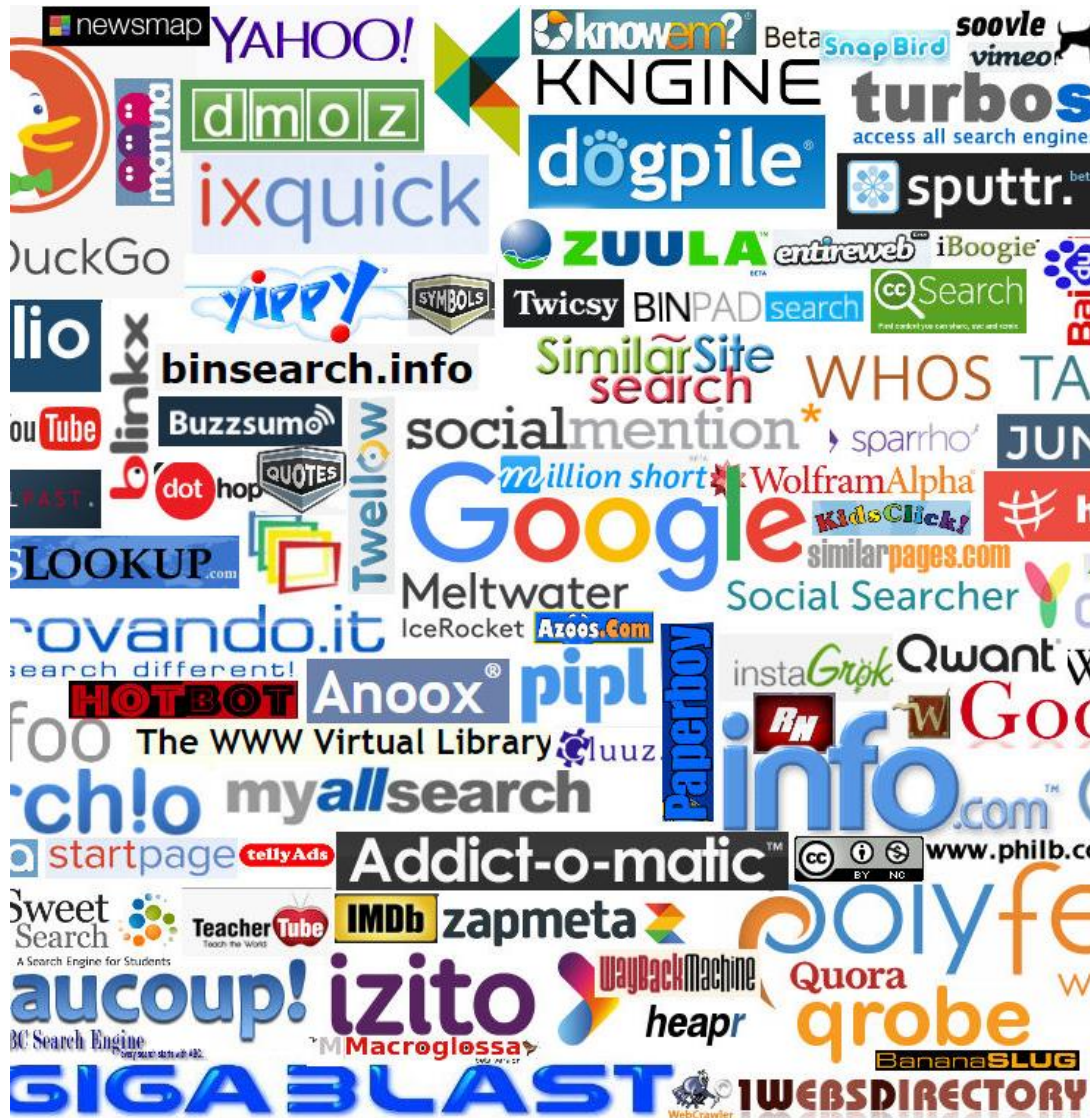
- ❖ Documents with the highest frequencies of the search TERM relative to their total length are selected.

# Potential algorithms

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## RETRIEVAL MODEL

- ❖ Probabilistic (statistical) models
- ❖ Imagine we have 10 documents in our COLLECTION.
- ❖ Each time we receive a QUERY we show the 5 random DOCUMENTS but each time the order of their appearance is randomised.
- ❖ *A priori*, what is the expected probability of each document being clicked on?
- ❖ Suppose we conducted an experiment with 10 users and each user used the same QUERY. We notice that 5 out of 10 users selected document 1. Is this what we would expect?
- ❖ Would the position a document is shown likely affect the probability of a user clicking on it?



(Bradley, 2016)

# Search Engines

- ❖ Web Search Engines (SE) are the most visible / well-known IR applications.
- ❖ They are among the most used and largest IR systems on the planet.
- ❖ Rather than return documents, search engines output an ordered list of HTML links (URLs) ranked according to their relevance.

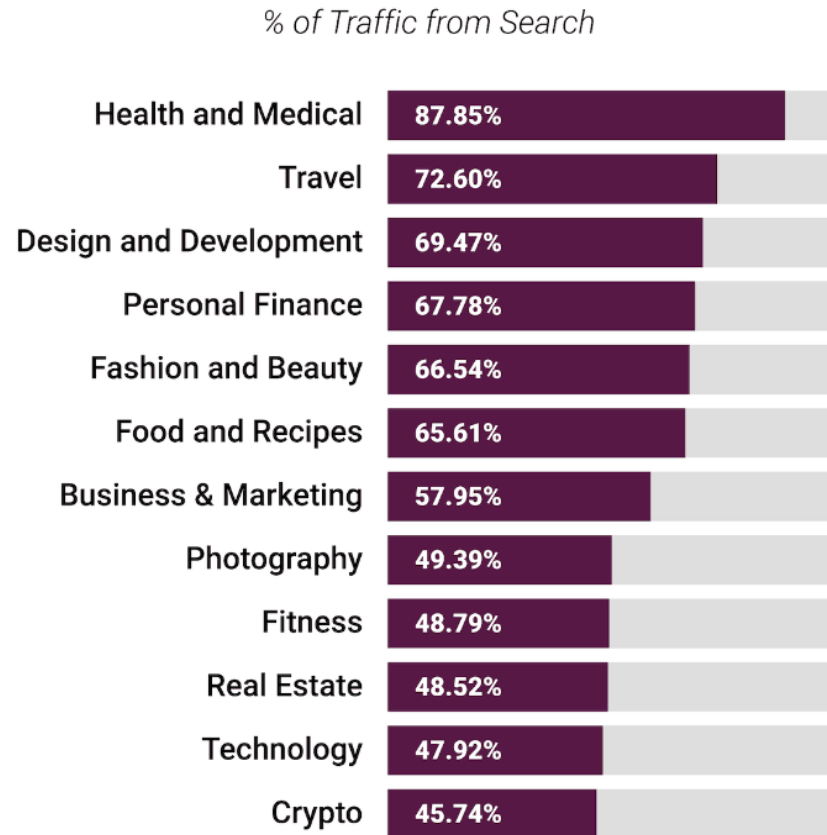
# Purpose of search engines

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- ❖ The web is growing at an enormous rate, roughly doubling in size every 3 years.
- ❖ Search engines act as **FILTERS** for the vast amount of context available online.
- ❖ If relevant web content is not easily accessible it can lead to INFORMATION OVERLOAD. Whereby we are unable to make effective decisions because there is too much information for us to efficiently process.
- ❖ Search engines rank 2<sup>nd</sup> – behind email, as the most frequently used internet application.
- ❖ We study them because users of our site will most likely find us through SEs – having a knowledge of how they work is critical.

# Proportion of traffic from search (organic and paid)

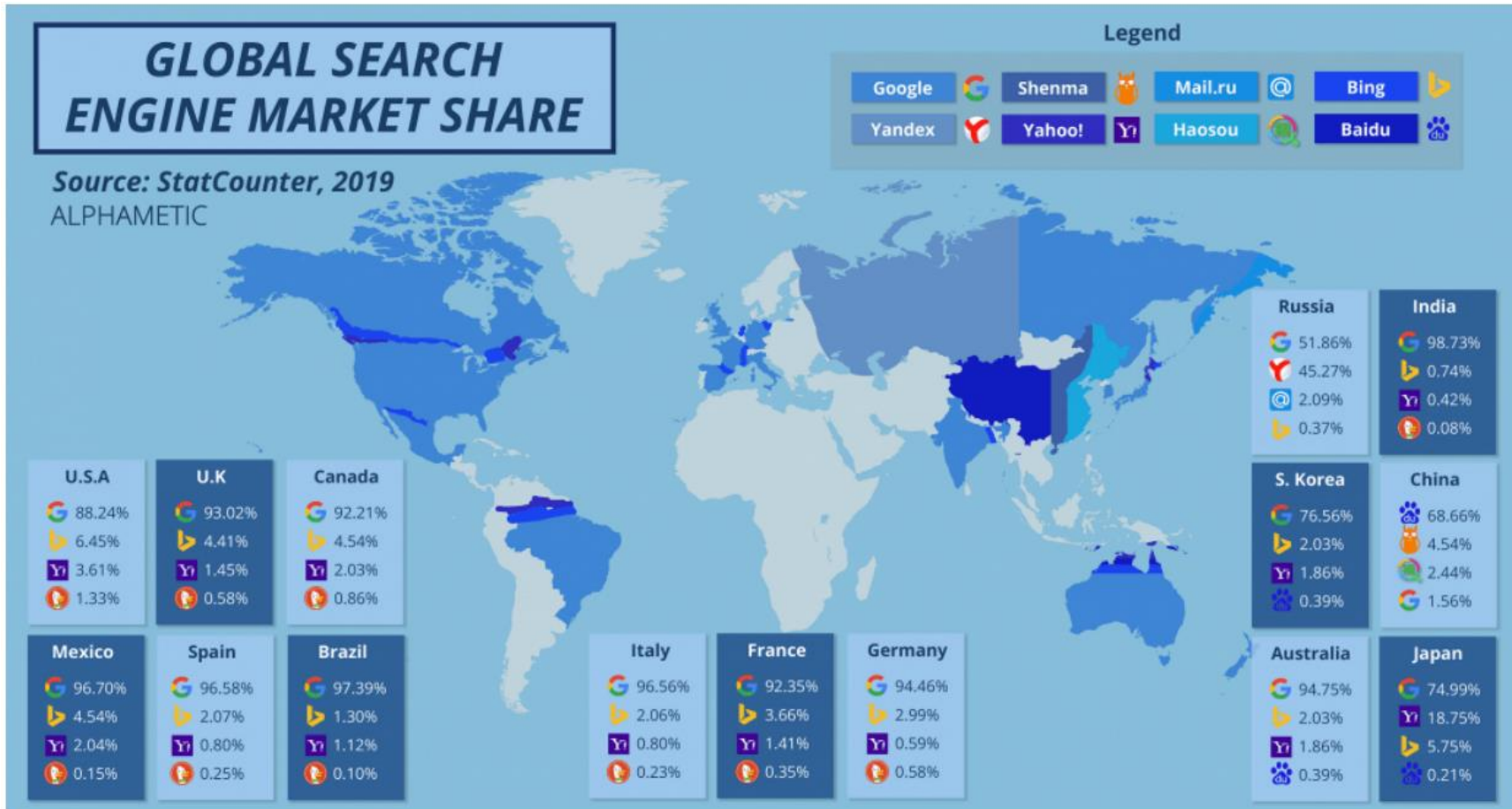
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(GrowthBadger, 2020)



# Search engines around the world today





# The search engine market

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Today's search engine (SE) market is mostly dominated by a few big names although several smaller players exist.

- ❖ This can partially be explained by (1) the capital-intensive nature of providing a search engine service, the (2) integration of search with other internet products and (3) decades of IP development/refinement.

## **META-SEARCH ENGINES**

Some search engines, known as meta-search engines, provide their own results based on a mixture of the results of some of the others (DuckDuckGo). While originally most search engines were part of some form of web portal, nearly all now use the simplified Google "inspired" layout.

# How does a SE work?

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Which one of the following two statements do you think it is closer to truth?

**A.** When a user submits a query to a SE, the SE starts searching the web to identify documents that could answer the specific query.

**OR**

**B.** When a new page / document is added on the web, (submitted to a SE) a SE analyses this document and extracts information in order to use it if a relevant query will come up.

# Key web search engine components

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Search engines (as information retrieval systems in their own right), share many of the same components, with a couple of key differences.

The generalised architecture of a SE includes:

- Query Input
- Query pre-processing
- A retrieval model (handling results ranking and merging)
- An Index (and supplemental Index)
- A database of web documents or fragments (for search previews)
- Web crawler (or search engine spider)

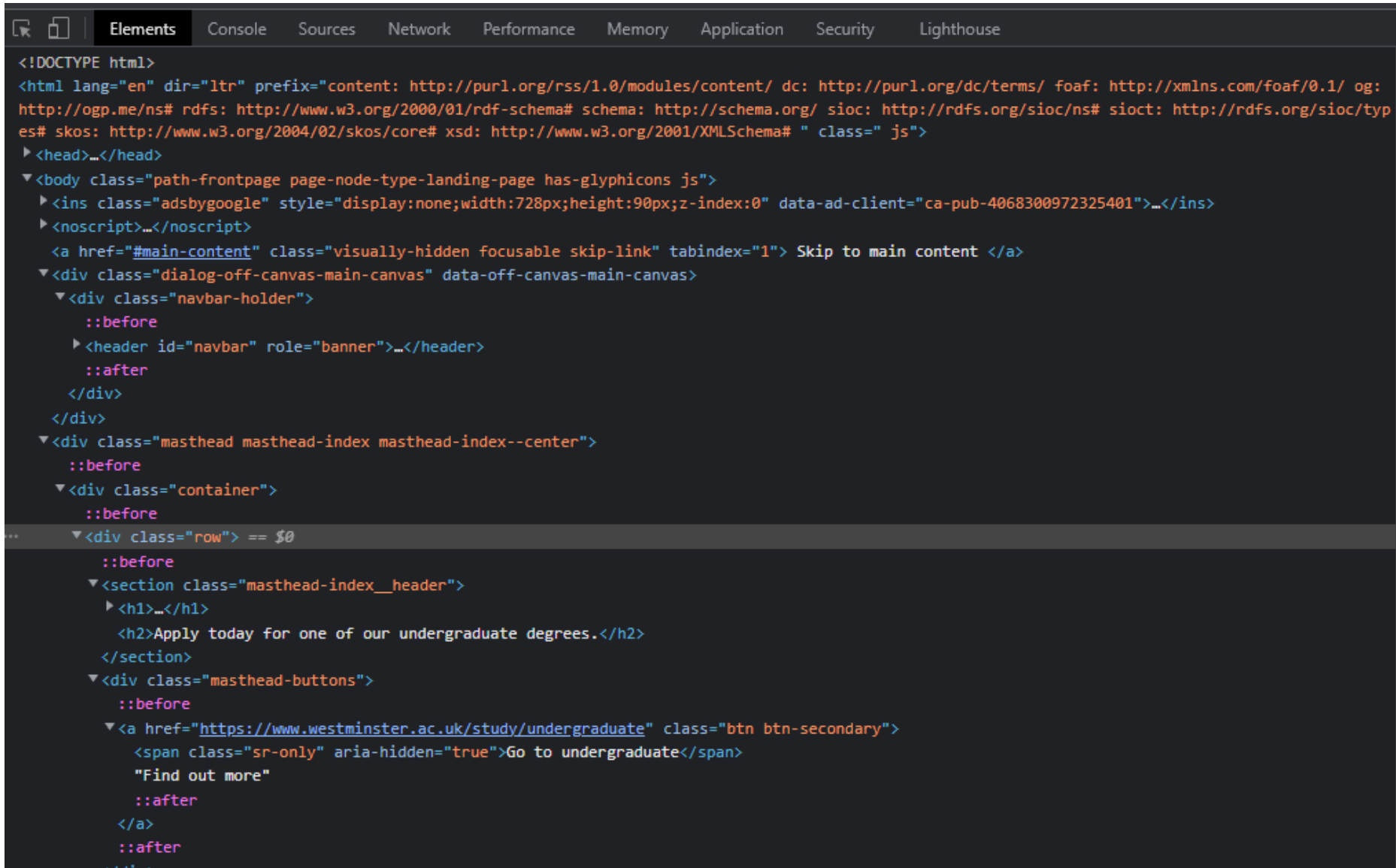
**Which of these information system components do you believe is the most challenging to design and implement?**

# The role of the crawler

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***Search Engines use web crawlers (spiders) to index websites.***

- ❖ When a website page is submitted to a search engine (by completing their required submission page), the search engine spider will **index** the entire site.
- ❖ A 'spider' is an automated program that is run by the search engine system. A Spider visits a web site, reads the content on the actual site, the site's Meta tags and also follows the internal links through the site.
- ❖ The spider then returns all that information back to a central repository, where the data is indexed. It will follow each link that it finds on each page and add them to the queue of sites to visit later.



```
<!DOCTYPE html>
<html lang="en" dir="ltr" prefix="content: http://purl.org/rss/1.0/modules/content/ dc: http://purl.org/dc/terms/ foaf: http://xmlns.com/foaf/0.1/ og:
http://ogp.me/ns# rdfs: http://www.w3.org/2000/01/rdf-schema# schema: http://schema.org/ sioc: http://rdfs.org/sioc/ns# sioc: http://rdfs.org/sioc/typ
es# skos: http://www.w3.org/2004/02/skos/core# xsd: http://www.w3.org/2001/XMLSchema# " class=" js">
  <head>_</head>
  <body class="path-frontpage page-node-type-landing-page has-glyphicons js">
    <ins class="adsbygoogle" style="display:none;width:728px;height:90px;z-index:0" data-ad-client="ca-pub-4068300972325401">_</ins>
    <noscript>_</noscript>
    <a href="#main-content" class="visually-hidden focusable skip-link" tabindex="1"> Skip to main content </a>
    <div class="dialog-off-canvas-main-canvas" data-off-canvas-main-canvas>
      <div class="navbar-holder">
        ::before
        <header id="navbar" role="banner">_</header>
        ::after
      </div>
    </div>
    <div class="masthead masthead-index masthead-index--center">
      ::before
      <div class="container">
        ::before
        <div class="row"> == $0
          ::before
          <section class="masthead-index__header">
            <h1>_</h1>
            <h2>Apply today for one of our undergraduate degrees.</h2>
          </section>
          <div class="masthead-buttons">
            ::before
            <a href="https://www.westminster.ac.uk/study/undergraduate" class="btn btn-secondary">
              <span class="sr-only" aria-hidden="true">Go to undergraduate</span>
              "Find out more"
            </a>
            ::after
          </div>
        </div>
      </div>
    </div>
  </body>
</html>
```

In the Chrome browser, we can press F12 to access the developer console and view the HTML code from the perspective of the “Spider”

# Updating the index

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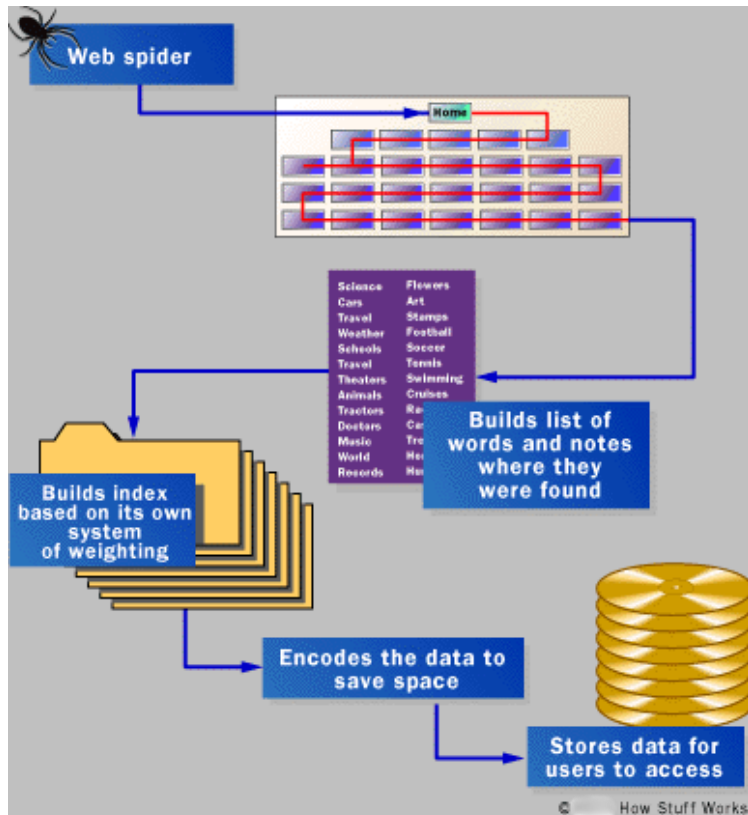
The spider will periodically return to the sites to check for any information that has changed.

The frequency with which this happens is determined by the moderators of the search engine.

The update frequency may also depend on:

- ❖ How old the site is. (why?)
- ❖ The niche or category of the website. (which types of websites might need updating more often?)

# Graphically



from: <http://computer.howstuffworks.com/internet/basics/search-engine1.htm>

- ❖ In practice, search engines will distribute the INDEX and DATABASE of web pages across multiple computers using a process known as **sharding**.
- ❖ When a new QUERY is received, the QUERY is broadcast to a group of shards. Each **shard** sends back matching documents (a result set) which are then merged together and ranked.

# Retrieval models

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- ❖ Once pages are in the INDEX it remains to **rank and order** them when a QUERY arrives.
- ❖ To increase **performance**, some pre-ordering and ranking is performed before QUERIES are matched against the INDEX (Optimisation)
- ❖ Similarly, the INDEX might be **pruned** to increase the quality of INDEX by removing sites associated with spam, illegal activity, duplicate content, thin-content or attempted gaming of the search results.
- ❖ Like Coca-Cola, the exact formula search engines use to calculate relevance of a web page to a QUERY are **closely guarded secrets**.
- ❖ We have already seen examples of retrieval models from IR theory, in practice SE retrieval models are more complex.





(Wired, 1998)

## PageRank™- Key extensions to retrieval models (1)

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- ❖ One of the most important improvements to search engine retrieval models was brought about by the development of the PageRank™ algorithm.
- ❖ The PageRank™ algorithm was developed by Google Engineers (Larry Page [L] and Sergey Brin [R]) and was published as an academic paper in January 1998.

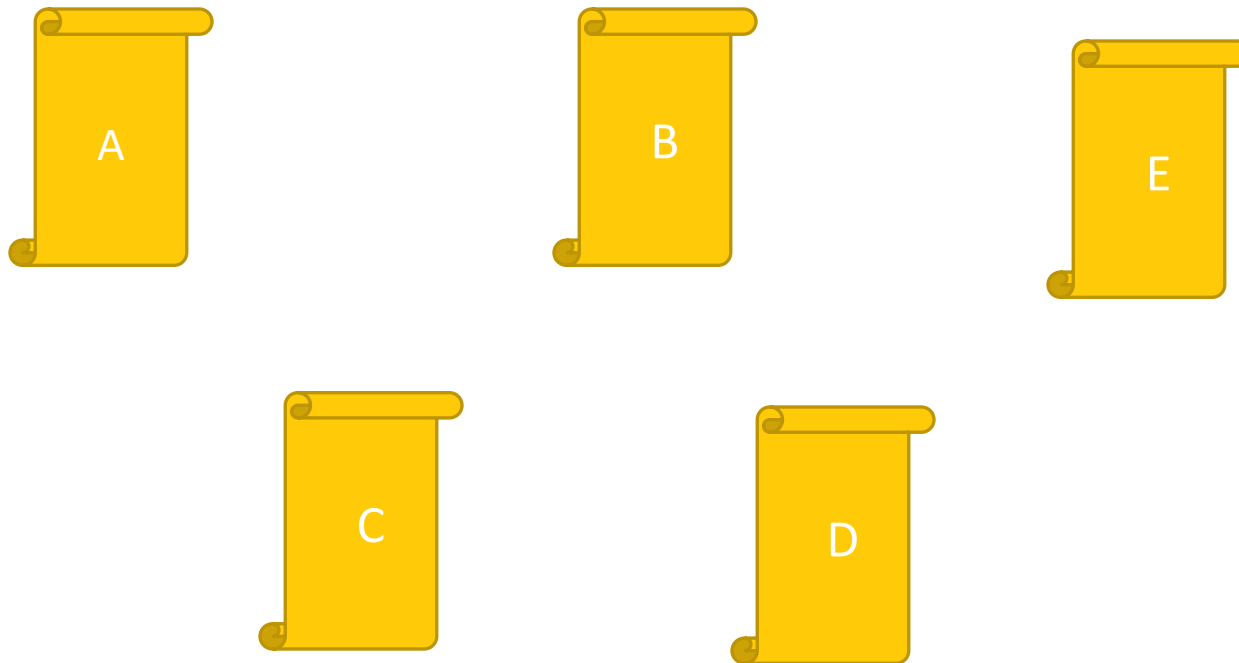
# PageRank – working principles

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- ❖ The relevance of a web page to a search query is SUBJECTIVE
- ❖ It is possible to supplement the SUBJECTIVE relevance of a page with an OBJECTIVE measure of its quality.
- ❖ Each page on the web contains a number of OUTBOUND links to other pages and websites.
- ❖ Each website therefore has an arbitrary number of INBOUND links pointing to its home page or any one of its sub-pages.
- ❖ Since links are created manually, this implies that a website has some confidence or expectation that another site is relevant or of value.
- ❖ The greater the number of INBOUND links to a website, the greater the confidence in that website by the collective web as a whole.

# PageRank – example 1

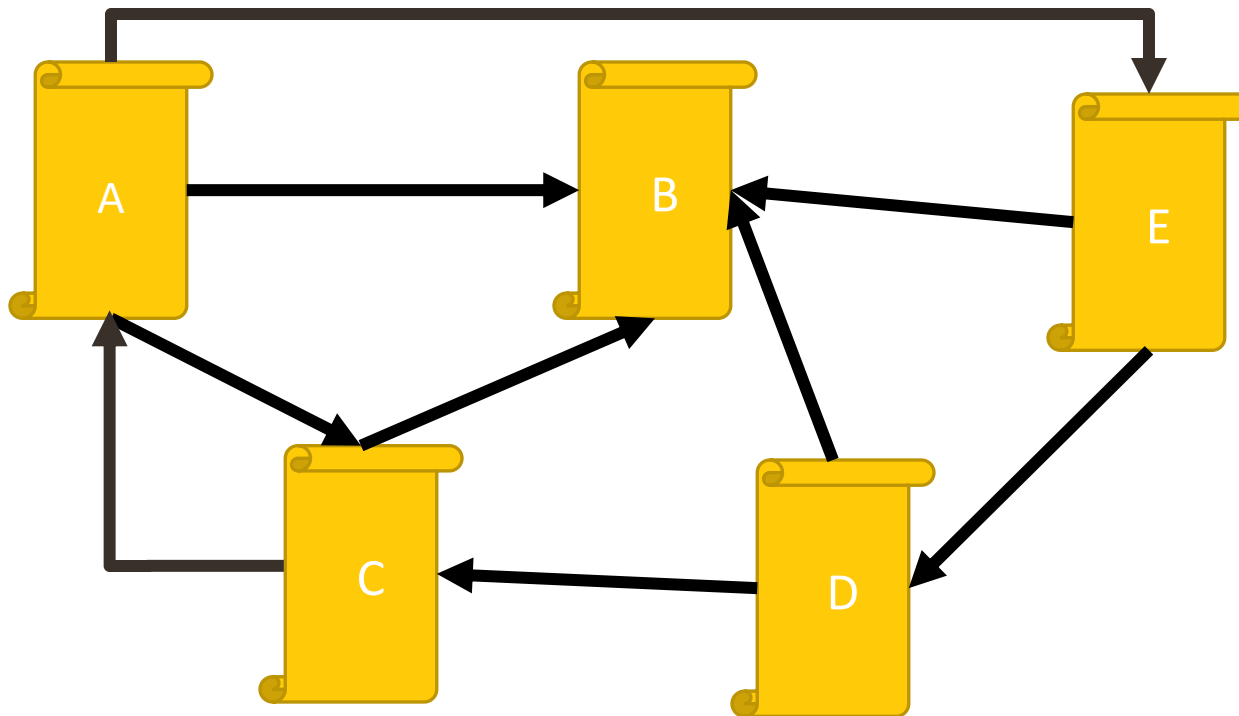
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Gather together the list of pages/websites for consideration (spider)

# PageRank – example 2

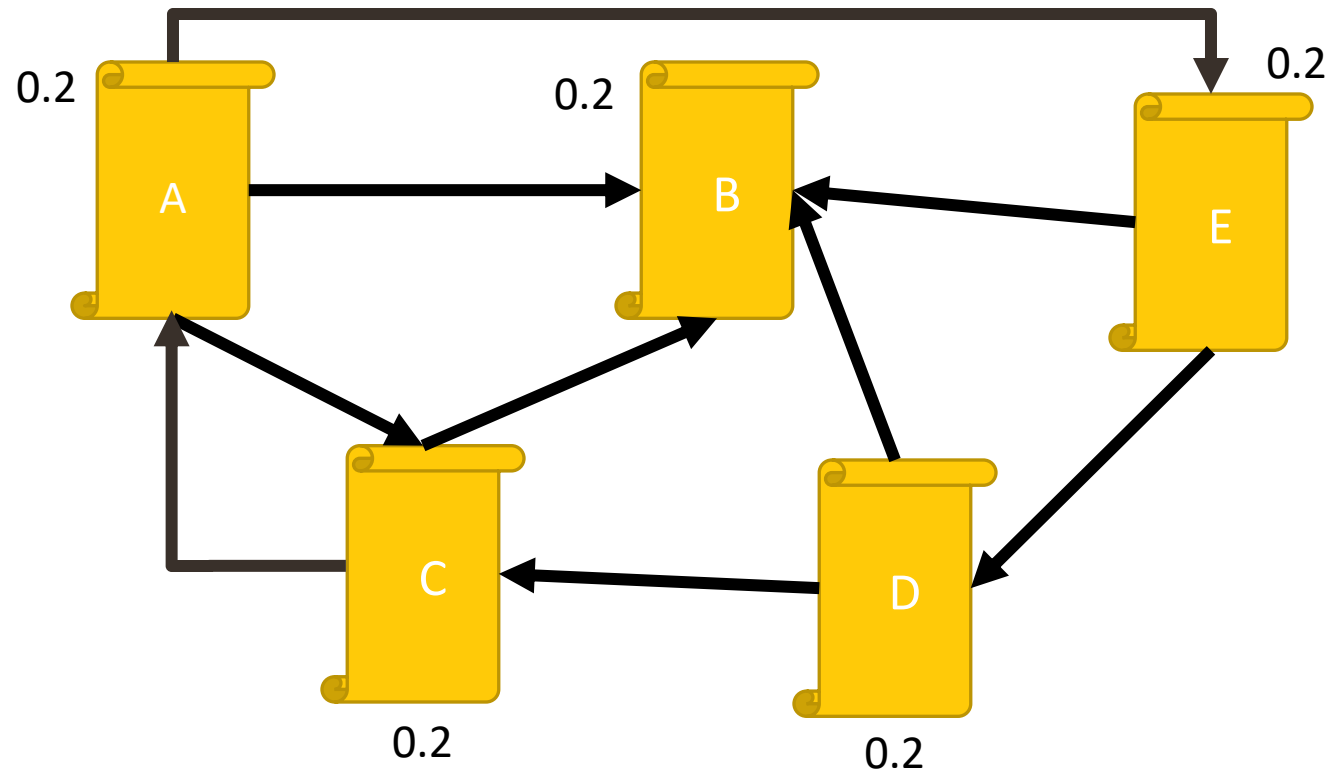
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Record the links made between each website/page

# PageRank – example 3

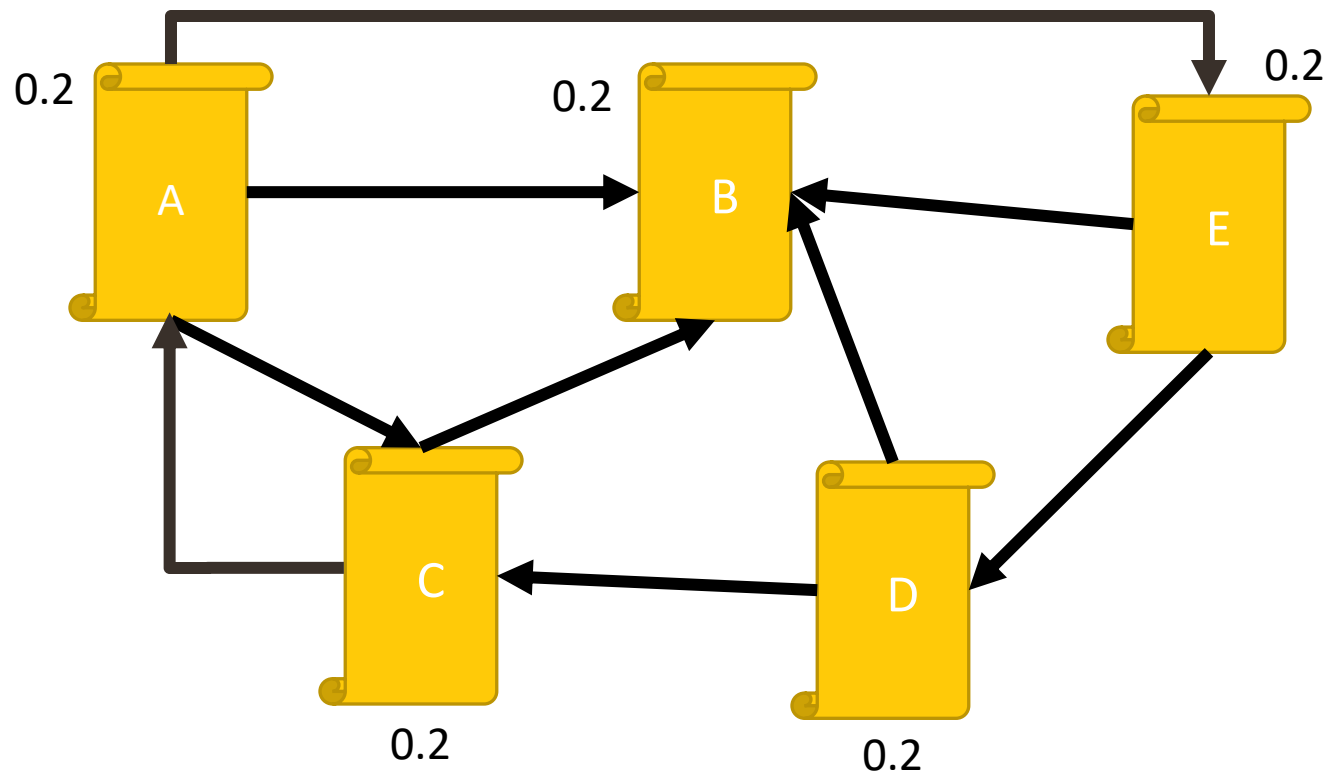
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Assign an initial PageRank™ of  $1/N$  to each website/page i.e.  $\sum_i^N Pr_i = 1$

# PageRank – example 4

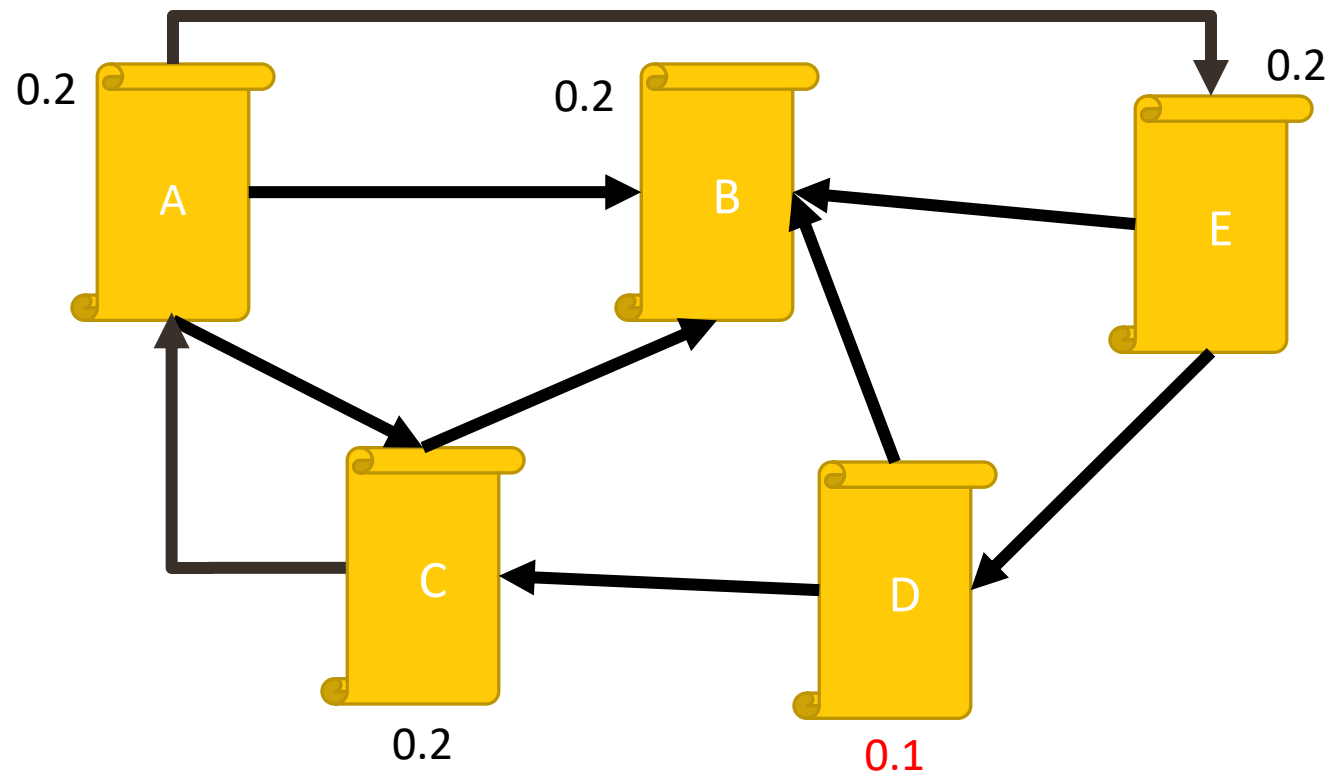
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Iteratively update each website/pages PageRank™ by summing a proportion of the PageRank from any pages pointing to it.

# PageRank – example 5

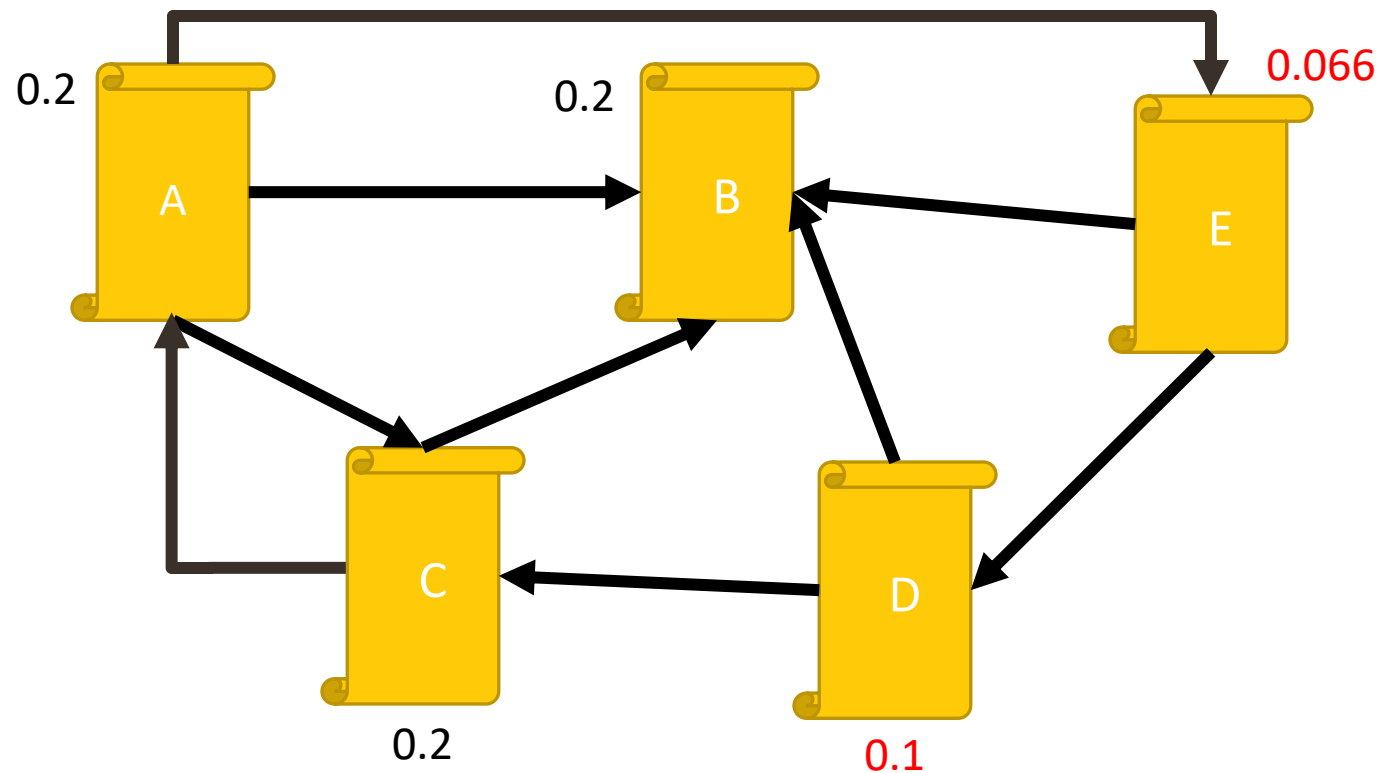
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$$\text{PageRank}(D) = \text{PageRank}(E)/2 = 0.2 / 2 = 0.1$$

# PageRank – example 6

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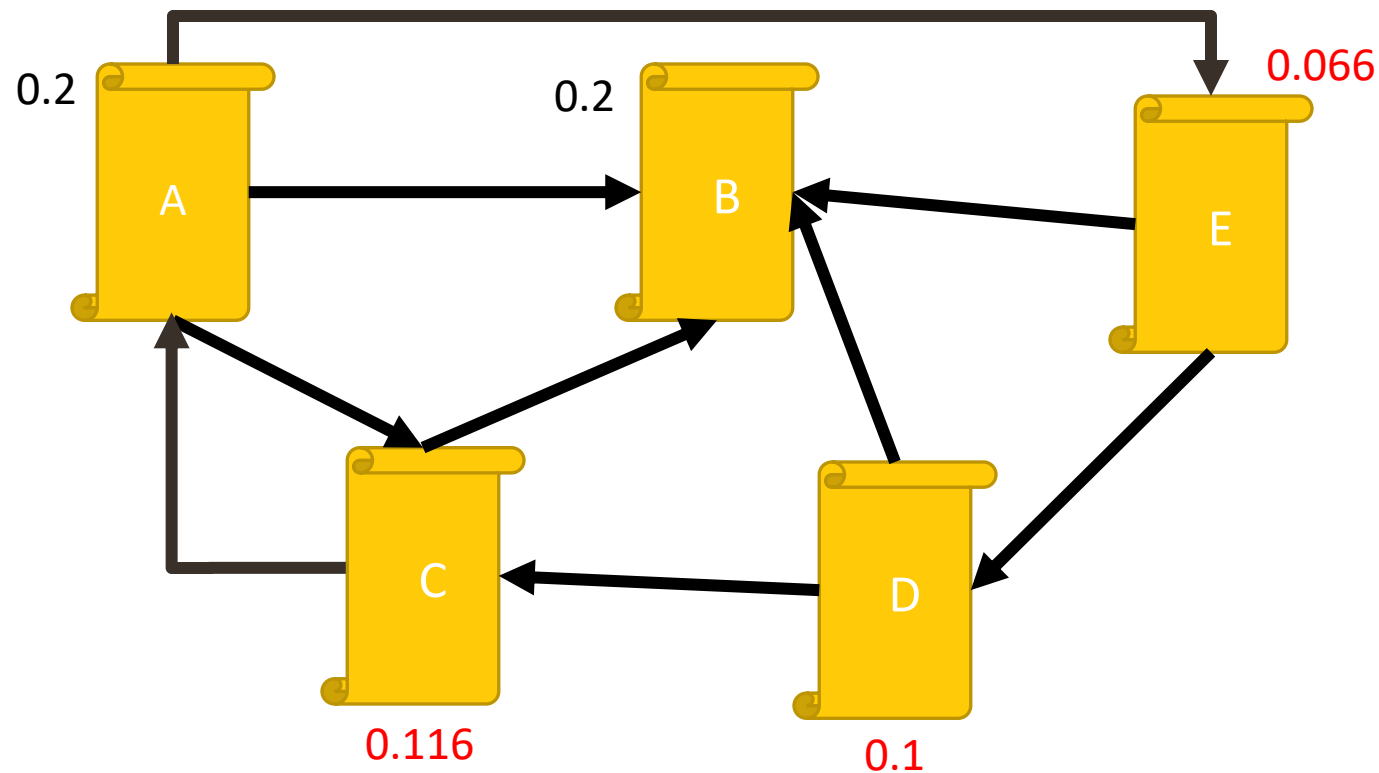


$$\text{PageRank}(E) = \text{PageRank}(A) / 3 = 0.066$$



# PageRank – example 7

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$$\text{PageRank}(C) = \text{PageRank}(D) / 2 + \text{PageRank}(A) / 3 = 0.05 + 0.066$$

# PageRank

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- ❖ The PageRank of a page depends on the PageRank of all other pages in the network.
- ❖ After many thousands of iterations, the PageRank values of each page begin to stabilise close to their true values.
- ❖ The PageRank algorithm is fairly difficult to “game” and is somewhat self-policing since it depends on links being created from either:
  - ❖ A large number of websites with a low PageRank
  - ❖ A small number of websites with a high PageRank

# Semantics and language - Key extensions to retrieval models (2)

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Language presents a **challenge** for search engines because it is difficult to represent the intricacies and semantics of speech in a binary system.

- Syntax (HOW we say something)
  - Colloquialisms
  - Sentence structure and order
  - Abbreviations and short hands
- Semantic (MEANING of what we say i.e. Hungry -> Food -> Take Away)
- Pragmatics and intention (**RELATIVE meaning**: The interpretation of a phrase/sentence is a largely inferential process, i.e. tax rules -> punishment for not paying tax)

# Pragmatics – Little test

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**Are you able to read this sentence? Most computers find this very challenging but studies have shown humans can easily do it.**

This is because humans use their own knowledge and experience to interpret text.

# Semantics and language - Key extensions to retrieval models (2)

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A simple test of a search engine proposed by Barney Pell, previously CEO of powerset, a company that went on to develop a natural language search engine.

Consider the **TWO** search queries:

“book by children”

“book about children”

Do we agree that these two search queries are fundamentally different and we would expect mutually exclusive sets of results?

**What happens in practice?**

# Semantics and language - Key extensions to retrieval models (2)

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Natural Language (NL) processing is a growing field of computer science that provides tools and methodologies for working with text. Many of these advancements are incorporated into modern search engines.

- ❖ Word stemming and lemmatisation (identify the core meaning of a word)
- ❖ Parts of Speech tagging (to identify the role of each word in a query)
- ❖ Named entity recognition (to identify people or places)
- ❖ Semantic tables and ontologies (identifying other words that have the same meaning or implication through WordNet)
- ❖ NLTK is a popular natural language toolkit for Python

# In Summary

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- ❑ We have reviewed some important principles of the web and the evolution of search on the web.
- ❑ Search engines provide an ability to query the web and identify relevant web pages or content.
- ❑ Search engines, as information retrieval systems themselves, borrow many of the core concepts and components of IR systems.
- ❑ Search engines are responsible for circa 40-80% of traffic to modern websites, thus they play a key role in driving acquisition
- ❑ Search engines use complex retrieval models, the most famous of which involves the development of the PageRank algorithm in 1998.
- ❑ Next week we'll present and discuss the most common techniques to optimize a search engine in order to make a web site more visible.